Again this number is not a measure or necessarily such a good indicator, of the instantaneous rate v at which the runner is moving 0.5 h into the race. If we determine that rate at 0.6 h the runner is 5.7 km from the starting line, then the average velocity from 0 h to 0.6 h is

$$v_{ave} = \frac{5.7-5}{0.6-0.5} = 7 km/h$$



Gottfried Wilhelm Leibnitz
(1646 - 1716)

The latter number is a more realistic measure of the rate v. By "shrinking" the time interval between 0.5 h and the time that corresponds to a measured position close to 5 km, we expect to obtain even better approximations to the runner's velocity at time 0.5 h.

This problem of finding velocities leads us to deal with the general problem of finding the derivative of a general mathematical model represented by the analytic equation, $y = f(x)$ Consequently, we will move towards in achieving the following objectives and subsequently deal with the analysis of derivatives.

Learning Objectives

On completion of this chapter, the students are expected to

- acquire the concept of a derivative as limit of quotients.
- visualise the concept of derivative geometrically.
- understand derivative as a process of measuring changes.
- realise derivative as a tool to measure slopes of tangents to curves / rates of changes.
- understand different methods of differentiation.
- apply calculus as a tool to solve everyday real life problems.

10.2 The concept of derivative

Calculus grew out of four major problems that mathematicians were working on during the seventeenth century.

(1) The tangent line problem

(2) The velocity and acceleration problem

(3) The minimum and maximum problem

(4) The area problem

We take up the above problems 1 and 2 for discussion in this chapter while the last two problems are dealt with in the later chapters.

10.2.1 The tangent line problem

What does it mean to say that a line is tangent to a curve at a point? For a circle, the tangent line at a point P is the line that is perpendicular to the radial line at a point P, as shown in fig. 10.1.

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